Dataset Construction Methodologies for FFR-GAN Project

A Comprehensive Approach to Building Datasets for Patient-Specific Fractional Flow Reserve Prediction Using Generative Adversarial Networks

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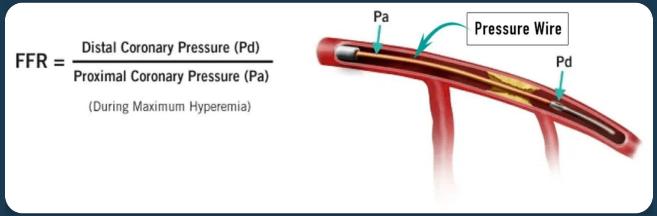


The Challenge: Patient-Specific FFR Prediction

Current Limitation

Existing CNN models predict FFR from anatomical features extracted from coronary CT angiography, achieving diagnostic accuracy of 75-85%. However, they fail to account for patient-specific physiological factors.

Missing Patient Context

Age, gender, diabetes, hypertension, and obesity significantly regulate coronary hemodynamics and microvascular function, yet are ignored by anatomy-only models.

Multicentric Variability

Inter-site differences in imaging protocols, scanner types, and patient populations affect FFR predictions but remain unaddressed in current approaches.

Our GAN-Based Solution

We propose a conditional GAN that refines baseline CNN predictions by integrating patient-specific conditioning variables, enabling personalized FFR adjustment.

1. **Baseline CNN Prediction**
   Predicts FFR from 28-dimensional anatomical features (vessel geometry, stenosis severity, plaque characteristics)
2. **Patient-Specific Conditioning**
   Incorporates demographics, comorbidities, and multicentric identifiers as conditioning variables
3. **GAN Refinement**
   Generator adjusts FFR based on patient context; Discriminator ensures physiological plausibility against real measurements

Approach 1: Synthetic Data Generation

Algorithmic generation of synthetic coronary artery trees with computational fluid dynamics (CFD) simulation for ground truth FFR labels.

Scale

10,000-20,000 cases with dense anatomical parameter space coverage, this makes the dataset suitable for training DL models.

Labeling

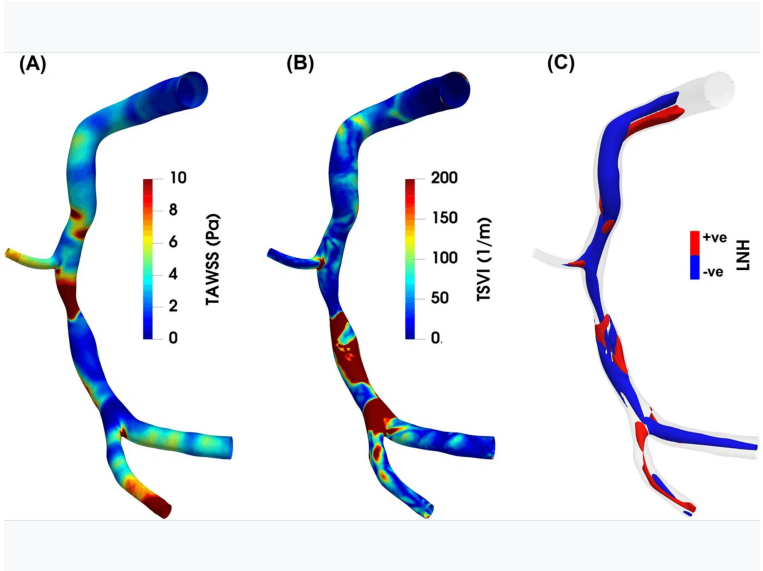
CFD-computed FFR at all centerline locations, which is deterministic and reproducible

Advantages

No privacy concerns, systematic exploration of rare anatomies, large-scale training

Limitations

Simplified hemodynamics, may not capture full clinical complexity and imaging artifacts



Computational Fluid Dynamics for Synthetic Data

CFD simulation generates FFR values across synthetic coronary trees by solving Navier-Stokes equations under hyperemic boundary conditions, providing deterministic ground truth labels for large-scale training.

Approach 2: Clinical Trial Data Collection

Collection of real patient CCTA images, invasive FFR measurements, and patient-specific variables from multicenter prospective trials. This approach captures authentic anatomical complexity, inter-patient variability, and multicentric differences that synthetic models cannot fully replicate.

Scale

400-600 vessels from multiple clinical centers with diverse patient populations. Smaller sample size compared to synthetic data but represents real-world clinical heterogeneity.

Labeling

Invasive FFR measurements (gold standard) obtained during cardiac catheterization, and PET-derived myocardial blood flow providing complementary physiological assessment.

Advantages

Authentic anatomical complexity including rare variants, real imaging artifacts from clinical scanners, true patient variability in hemodynamics, and multicentric representation for generalizability.

Limitations

Expensive and time-consuming data collection process, strict privacy regulations, limited sample size restricting statistical power, and selection bias toward patients clinically indicated for invasive procedures.

Representative Multicenter Clinical Trials

NXT Trial: Prospective multicenter study with 484 vessels providing CCTA images, invasive FFR measurements, and comprehensive patient demographics across multiple clinical

Recommended Hybrid Data Strategy: Overview

Rationale: Combine the scalability of synthetic data with the authenticity of clinical measurements to train a robust baseline model and a patient-specific refinement layer.

1

STAGE

Synthetic Data Generation for Baseline CNN Training

Generate 10,000-20,000 synthetic coronary artery trees using algorithmic methods. Compute FFR labels via CFD simulation. Train baseline CNN to learn fundamental anatomy-to-hemodynamics relationships.

- Large-scale training dataset with broad anatomical coverage
- Deterministic CFD labels ensure reproducibility
- No privacy concerns
- Baseline CNN achieves $R^2 \approx 0.85$-0.90

Recommended Hybrid Data Strategy: Clinical Integration

2
STAGE

Multicenter Clinical Data Collection

Collect 200-300 vessels from 3-5 clinical centers with CCTA images, invasive FFR measurements, and comprehensive patient-specific variables (demographics, comorbidities, medications, site identifiers).

- Real anatomical complexity and imaging artifacts
- Gold standard invasive FFR labels
- Patient-specific conditioning variables captured
- Multicentric variability naturally represented

3
STAGE

Conditional GAN Training for Patient-Specific Adjustment

Train conditional GAN using clinical dataset. Generator refines baseline CNN predictions by incorporating patient characteristics. Discriminator ensures physiological plausibility against real FFR measurements.

- Learns patient-specific modulation of FFR
- Multi-component loss: adversarial + reconstruction + physiological
- Expected 10-20% MAE reduction over baseline
- Leave-one-center-out validation for generalizability

Key Advantage of Hybrid Approach

Data Requirements: Input Features

Baseline CNN Input Specification: The model requires comprehensive anatomical and plaque-based features extracted from coronary CT angiography to predict FFR values at each centerline location.

Anatomical Features (28D)

- Lumen radius at centerline location
  Actual vessel diameter at measurement point
- Reference radius
  Expected healthy vessel diameter without stenosis
- Ischemic weight
  Quantifies downstream myocardial territory perfused
- Upstream stenosis parameters (×4)
  Proximal radius, minimum radius, stenosis length for four most significant upstream lesions
- Downstream stenosis parameters
  Characterization of distal lesions affecting flow distribution
- Bifurcation geometry
  Branching angles and vessel diameter ratios at bifurcations

Plaque Characteristics

- Total plaque volume
  Overall atherosclerotic burden in vessel segment
- Calcified plaque volume
  Dense calcium deposits visible on CT
- Non-calcified plaque volume
  Soft plaque components without calcification
- Low-density non-calcified plaque
  Lipid-rich vulnerable plaque with greatest FFR impact
- Stenosis severity (% diameter)
  Percent reduction in lumen diameter at narrowest point

Data Requirements: Conditioning Variables & Labels

Patient-Specific Variables

Demographics

→ Age (continuous: 40-85 years)

→ gender (binary: male/female)

Clinical Comorbidities

→ Diabetes mellitus (binary)

→ Hypertension (binary)

→ Obesity (BMI: 18-40 kg/m²)

→ Prior cardiovascular disease

→ Current smoking status

→ Hyperlipidemia

Multicentric Identifiers

→ Clinical center ID (categorical)

→ Scanner type and protocol

→ Imaging acquisition date

Target Labels

Primary: Invasive FFR

Measured during cardiac catheterization using pressure wire advanced distal to stenosis under maximal hyperemia (adenosine administration).

Clinical threshold: FFR ≤ 0.80 indicates hemodynamically significant stenosis requiring revascularization

Synthetic: CFD-Computed FFR

Generated via computational fluid dynamics simulation on synthetic coronary trees. Achieves 75-85% diagnostic accuracy compared to invasive FFR.

Complementary: PET MBF

Quantitative myocardial blood flow from positron emission tomography. Captures both epicardial stenosis and microvascular function.

Ischemia threshold: MBF ≤ 2.30 mL/min per gram indicates reduced perfusion

Label Format

→ Continuous regression target (0.0-1.0)

→ Binary classification (ischemic vs. non-ischemic)

Hierarchical Data Organization

LEVEL 1
Patient
Demographics, comorbidities, medications

→

LEVEL 2
Study
Imaging parameters, scanner type, date

→

LEVEL 3
Vessel
LAD, LCX, RCA anatomical features

→

LEVEL 4
Lesion
Stenosis features, FFR values

Relational Database Structure

Linked Tables: Patient table (patient_id, age, gender, diabetes, hypertension, BMI) → Study table (study_id, patient_id, scan_date, center_id) → Vessel table (vessel_id, study_id, vessel_type) → Lesion table (lesion_id, vessel_id, FFR, anatomical_features)

Enables efficient querying, maintains referential integrity, and supports complex analyses across hierarchical levels.

File Naming Conventions

Patient ID:	P01_0001
Study ID:	S_P01_0001_20240315
Vessel ID:	V_S_P01_0001_20240315_LAD
Lesion ID:	L_V_..._LAD_001

Ensures traceability from lesion-level predictions back to original patient data and imaging studies.

F
Findable
Unique identifiers, comprehensive metadata

A
Accessible
Standardized formats (DICOM, CSV, JSON)

I
Interoperable
Compatible with analysis software and tools

R
Reusable
Clear provenance, preprocessing documentation

Five-Phase Implementation Timeline: Data & Training

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Synthetic Data Generation

Weeks 1-3

- Implement coronary tree generator algorithm
- Generate 10,000-20,000 synthetic cases
- Assign geometric parameters from literature ranges
- Place stenoses with varying severity
- Run CFD simulations for FFR labels
- Generate patient-specific variables synthetically
- Export to CSV format with 28 features

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Baseline CNN Training

Week 4

- Prepare training/validation/test splits
- Normalize anatomical features
- Train 4-layer CNN architecture
- Implement batch normalization and dropout
- Evaluate on synthetic test set
- Target: $R^2 \approx 0.85$-0.90, MAE ≈ 0.03-0.04
- Save baseline model for GAN input

3

Clinical Data Collection

Weeks 5-12

- Establish data sharing agreements with 3-5 centers
- Obtain ethics approvals and consent forms
- Collect de-identified CCTA images
- Gather invasive FFR measurements
- Extract patient demographics and comorbidities
- Target: 200-300 vessels across centers
- Perform quality control and validation

Five-Phase Implementation Timeline: GAN & Validation

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GAN Training

Weeks 13-15

- Implement conditional GAN architecture
- Generator: baseline FFR + patient vars → adjusted FFR
- Discriminator: real vs generated patient-specific FFR
- Multi-component loss function
- Train for 100+ epochs with early stopping
- Target: 10-20% MAE reduction vs baseline

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Multicentric Validation

Weeks 16-17

- Leave-one-center-out cross-validation
- Evaluate generalization across sites
- Compare baseline CNN vs GAN-adjusted
- Analyze patient subgroup performance
- Generate publication-quality figures
- Document limitations and future work

Total Duration

17 Weeks

Key Deliverables

5 Milestones

Expected Improvement

10-20% MAE

Expected Performance Improvements

Baseline CNN Model

R^2 Score	0.85 - 0.90
Mean Absolute Error	0.03 - 0.04
Training Data	Synthetic
Patient Context	Not Integrated

GAN-Adjusted Model

R^2 Score	0.90 - 0.95
Mean Absolute Error	0.024 - 0.032
Training Data	Hybrid
Patient Context	Fully Integrated

10-20%

MAE Reduction

Expected Performance Gain: The GAN-adjusted model is anticipated to reduce mean absolute error by 10-20% compared to the baseline CNN through patient-specific conditioning. This translates to improved clinical accuracy in identifying hemodynamically significant stenosis, potentially reducing unnecessary invasive procedures and improving risk stratification for revascularization decisions.